

A Multi-Agent Orchestration Framework for Venture Capital Due Diligence

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Abstract

We present a fully automated multi-agent framework for corporate due diligence and market analysis in venture capital. The system runs on an event-driven orchestration architecture, combining Large Language Models (LLMs) with real-time web retrieval to synthesize unstructured data into structured investment intelligence. A central technical contribution is a programmatic extraction pipeline that reverse-engineers the frontend-to-backend communication of the Greek Business Registry (T.E.MH.), querying dynamic endpoints to retrieve official financial filings that are then parsed using a layout-aware OCR extractor. A structural fallback mechanism explicitly flags data absence rather than generating unverified figures, directly targeting hallucination in financial contexts. All workflow artifacts are publicly available to support replication.

Keywords: AI agents; Multi-Agent Systems; Workflow Automation; Due Diligence; Venture Capital; Retrieval-Augmented Generation; Hallucination Mitigation

1 Introduction

15 Evaluating prospective investments and monitoring portfolio companies demands the synthesis
of fragmented information across incompatible sources: unstructured web data, real-time
news, competitive analyses, and official financial registries that are often locked behind
administrative access controls. Manual desk research, the standard approach, introduces
human error, cognitive bias, and delays that compound across a portfolio (Gompers et al.,
20 2016; Kahneman, 2011).

LLMs offer powerful capabilities for natural language understanding and data synthesis
(Bommasani et al., 2021), but deploying them for financial due diligence exposes three
structural problems. First, training data cut-offs leave models blind to current market signals.
Second, LLMs generate plausible but factually incorrect information—a failure mode that
25 financial decision-making cannot tolerate (Ji et al., 2023). Third, base LLMs lack the ability
to autonomously query external databases, invoke APIs, or parse scanned financial documents;
frameworks that enable tool use and inter-agent communication are required to bridge this
gap (Q. Wu et al., 2023).

We address these problems with an event-driven multi-agent orchestration pipeline. Rather
30 than a single LLM prompt, the system decomposes due diligence into specialized sub-tasks
handled by distinct AI agents (Xi et al., 2023). Real-time search APIs supply current
market intelligence; a custom extraction module programmatically retrieves and parses official
financial filings from the Greek Business Registry (Γ.Ε.ΜΗ.).

This paper makes three contributions:

- 35 1. A modular, multi-agent architecture that automates end-to-end corporate research from
a single analyst trigger to a structured HTML report delivered by email.
2. A programmatic financial extraction pipeline that queries hidden Γ.Ε.ΜΗ. registry
endpoints, retrieves raw PDF documents, and applies layout-aware OCR to structure
financial data with preserved source citations.

3. A structural fallback mechanism that routes to open-source financial intelligence when registry data is unavailable, replacing potential LLM hallucinations with explicit, auditable data gaps. 40

2 Background and Related Work

2.1 Multi-Agent Frameworks

The field has evolved from single-prompt LLM interactions toward agentic workflows, where multiple LLM-powered agents each carry distinct roles, system prompts, and tool-use capabilities (Xi et al., 2023). Chaining agents allows each one to pass its verified output to the next, constructing reasoning chains that exceed the capacity of a single model call. Multi-step analytical problems requiring precision and iterative correction are the natural application domain for these architectures. 45 50

AutoGen operationalizes this paradigm through a conversational multi-agent framework in which agents negotiate task decomposition and tool invocation dynamically (Q. Wu et al., 2023). MetaGPT takes a complementary approach, assigning software-engineering roles—product manager, architect, engineer—to distinct agents coordinated via structured output schemas (Hong et al., 2024). LangGraph provides a graph-based state-machine abstraction for agent orchestration, enabling conditional branching and cycle detection in complex pipelines (LangChain Inc., 2024). Our system follows the same DAG-structured philosophy but is implemented on n8n (n8n, 2024), a low-code event-driven platform that exposes HTTP, conditional routing, and AI nodes without requiring custom agent code, substantially lowering the engineering barrier for non-technical investment teams. 55 60

2.2 LLMs for Financial Analysis

Applying agentic systems to finance is natural: general-purpose LLMs struggle with domain-specific vocabulary and numerical precision (Yang et al., 2023). BloombergGPT demonstrated

that domain-adapted pre-training yields measurable improvements on financial NLP benchmarks (S. Wu, Irsoy, et al., 2023). More recent multi-agent architectures have targeted specific financial tasks: MARAG-Fin distributes analytical subtasks across agents to improve trading decision accuracy and reduce hallucinations in volatile markets (Luckianto & Gunawan, 2026), while QuantAgents structures investment analysis as a simulated trading environment coordinated by specialist agents (Li et al., 2025). FinAgent extends this line by incorporating multimodal inputs—price charts, news sentiment, and fundamental data—into a unified agent pipeline (Zhang et al., 2024). Our work is distinguished from this body of literature by its focus on pre-investment due diligence rather than post-investment trading, and by its integration with an official national corporate registry that provides auditable, ground-truth financial data rather than market signals.

2.3 Retrieval-Augmented Generation and Real-Time Grounding

Dynamic information retrieval addresses the static knowledge problem: grounding model outputs in live search results and external data streams allows systems to bypass training cutoffs entirely (Lewis et al., 2020). For corporate intelligence, Open-Source Intelligence (OSINT) provides current market sizing, funding rounds, and competitor movements. Retrieval-Augmented Generation makes that intelligence auditable by linking outputs to primary sources (Gao et al., 2023).

2.4 Document Understanding and OCR

Automated document processing extends retrieval into non-digitized corporate records. Layout-aware parsers handle balance sheets and income statements without the structural loss typical of legacy OCR engines (Paruchuri, 2025). Combined with real-time web retrieval, these tools provide a path from fragmented market signals to formal regulatory filings within a single pipeline—the integration this work demonstrates.

3 System Architecture

The system runs on an event-driven automation platform (n8n, 2024), structured as a Directed Acyclic Graph (DAG) of processing nodes. A custom HTML form serves as the entry point 90 for investment teams: analysts select a target portfolio company and trigger the full research workflow with a single click, with the low-code backend hidden from view.

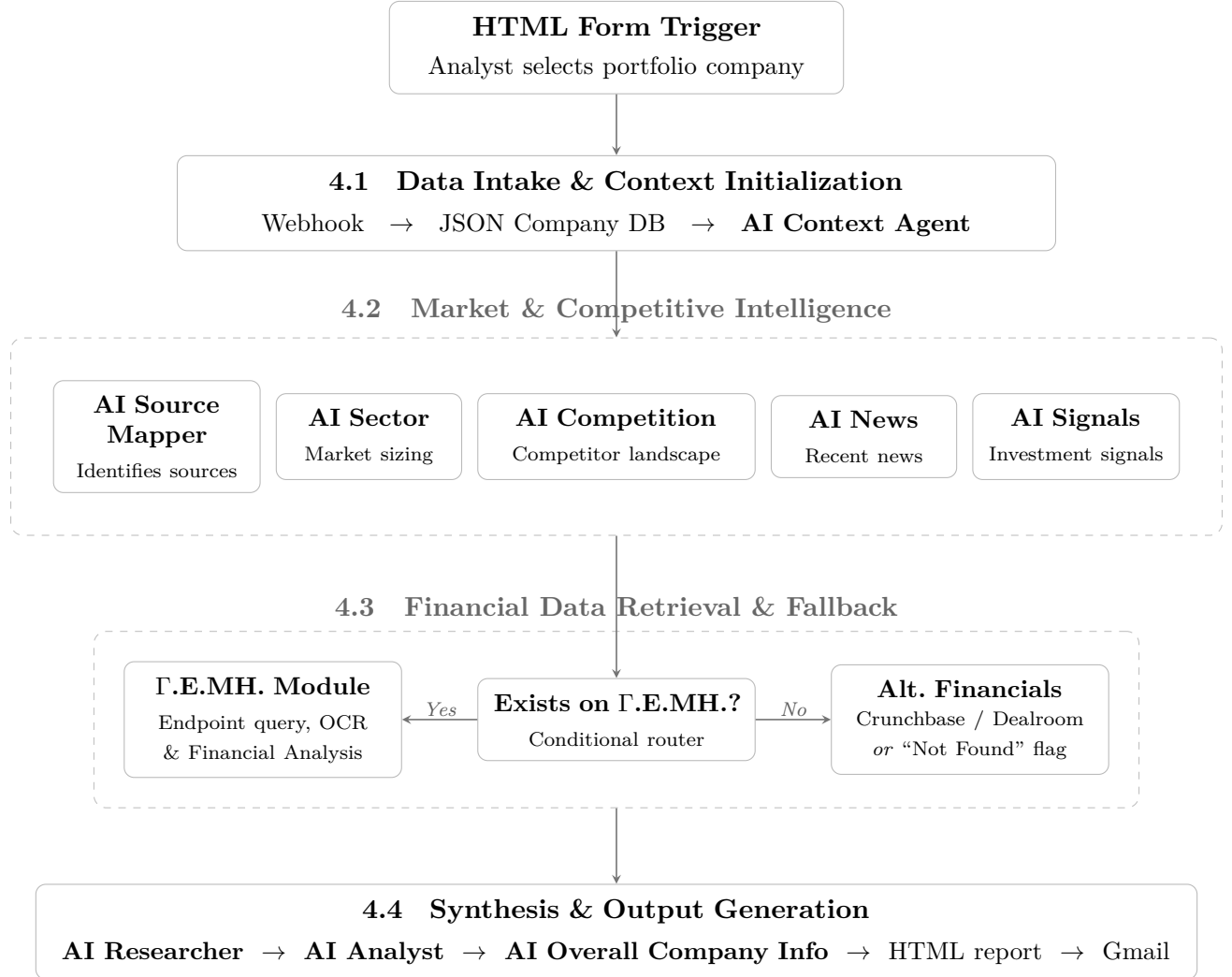


Figure 1: High-level overview of the event-driven multi-agent orchestration architecture, structured according to the four pipeline stages described in Sections 3–3.4.

3.1 Data Intake and Context Initialization

On submission, a Webhook node captures the payload. A transformation node maps the
95 selected company to a pre-scraped JSON database of baseline attributes: founders, sector, initial investment year, headquarters, and registration numbers. The **AI Context Agent** takes this structured input and produces a compact company profile that anchors all downstream research nodes.

3.2 Market and Competitive Intelligence Modules

100 Five specialized agents query the Perplexity Sonar Deep Research API (Perplexity AI, 2024) for real-time market intelligence. The **AI Source Mapper** runs first, identifying the most reliable sector-specific portals, databases, and news aggregators for the target company. The **AI Sector** and **AI Competition** agents then run in parallel against those sources, extracting market sizing, macro-environmental trends, and a competitive landscape that categorizes
105 competitors as direct, adjacent, or niche innovators. Once both complete, the **AI News** and **AI Signals** agents scan for recent strategic developments (partnerships, product launches) and extract quantifiable investment signals.

3.3 Financial Data Retrieval and Fallback Mechanisms

The most technically involved component interfaces with the Greek Business Registry
110 (Γ.Ε.ΜΗ.). Although the registry provides an official API, obtaining authenticated credentials involves administrative overhead. The framework instead reverse-engineers the portal’s frontend-to-backend communication, a method validated by prior open-source work targeting the same registry (Drakakis, 2025).

Registry data and corporate announcements are served as PDF documents loaded from
115 specific backend endpoints. A conditional router checks for a valid registry number (*arGEMI*). When one exists, an HTTP Request node mimics browser behavior, querying those endpoints

to retrieve the JSON payload of document IDs.

Document IDs are classified into two streams: **Corporate Modifications**—covering board changes, capital increases, and statutory modifications—and **Financial Statements**, comprising official balance sheets, income statements, and auditor reports. The pipeline 120 fetches raw PDFs, prioritizing the most recent fiscal years, and passes them through a layout-aware text extractor (Paruchuri, 2025). Two summarization agents, **AI FinSummary** and **AI ModSummary**, distill the extracted text into standardized financial metrics (Assets, Liabilities, Revenue, EBIT) with source citations preserved.

Preventing fabricated financial data is the system’s hardest constraint. When no Γ.E.MH.125 number is found—typically for companies incorporated abroad—the pipeline activates the **Alternative Financials** agent, which queries Crunchbase, Dealroom, and equivalent open-source databases. If those sources also return nothing, the agent writes a “Not Found” flag directly into the report, making the data gap explicit rather than filling it with generated figures. 130

3.4 Synthesis and Output Generation

Three synthesis agents combine the intelligence streams. The **AI Researcher** aggregates sector, competition, news, signals, and financial data into a strategic research note that surfaces recent developments and blind spots. The **AI Analyst** takes that note, produces an Executive Summary, assigns attractiveness scores across market timing and product 135 differentiation, and drafts 30-to-180-day recommendations for both the fund and the startup. The **AI Overall Company Info** agent generates a public-facing summary that founders can use to understand their external market footprint.

A formatting node converts the JSON outputs into a styled HTML report and delivers it to the requesting analyst through a Gmail API integration. 140

4 Report Structure and Fallback Behavior

The final artifact of the pipeline is a structured HTML report delivered by email. Its sections map directly to the agent layer that produced them, making the provenance of each claim traceable to a specific intelligence stream.

145 4.1 Report Anatomy

The report opens with a **Company Overview** produced by the **AI Overall Company Info** agent: a concise external-facing summary of the portfolio company’s market position, value proposition, and competitive footprint as reflected in publicly available signals. This section is intentionally written to be shareable with founders as a reflection of their external
150 presence.

The **Market Intelligence** block covers current market sizing and trajectory, macro-regulatory dynamics, and a timeline of recent strategic events (funding announcements, product launches, partnerships). Each claim carries an inline citation to the underlying Perplexity Sonar source retrieved by the **AI Sector** and **AI News** agents.

155 The **Competitive Landscape** section, contributed by the **AI Competition** agent, classifies identified competitors into three tiers—direct, adjacent, and niche innovators—with funding status, activity signals, and a positioning summary relative to the portfolio company.

The **Financial Summary** presents the structured line items extracted from F.E.MH. official filings: Revenue, Total Assets, Total Liabilities, and EBIT across available fiscal years,
160 each cited to the source PDF and page number extracted by the **AI FinSummary** agent. A **Corporate Events** timeline, produced by **AI ModSummary**, lists statutory changes such as capital increases and board amendments in chronological order. When registry data is unavailable, this block displays either a structured entry from Crunchbase or Dealroom, or an explicit “Not Found” flag (see Section 3.3).

165 The report closes with an **Analyst Assessment** from the **AI Analyst** agent: an

Executive Summary, attractiveness scores across market timing and product differentiation dimensions, and 30-to-180-day action recommendations addressed separately to the fund and to the portfolio company.

4.2 Fallback Behavior in Practice

The structural fallback is directly observable in the report output. When a company holds no 170 valid *arGEMI*—typically because it is incorporated abroad—the Financial Summary block displays a “Not Found” flag rather than any generated estimate. This flag is rendered with the same formatting as a populated entry, making the data gap explicit and immediately visible to the analyst. The design reflects a deliberate trade-off: a missing figure is less harmful to investment analysis than a plausible but unverifiable one. Analysts can therefore immediately 175 distinguish registry-sourced figures (auditable, official) from open-source approximations (Crunchbase, Dealroom) from explicit gaps—three distinct epistemic states that the system surfaces without conflating them.

5 Limitations

Several constraints bound the current system. First, the financial extraction component is 180 specific to Γ.E.MH., limiting registry-grade data to Greek-incorporated entities; companies registered abroad rely on open-source databases with lower coverage and less structured data. Second, the pipeline depends on three external commercial services—Perplexity Sonar, the LLM provider, and the Marker PDF extractor—making it subject to pricing changes, rate limits, and interface updates; a production deployment should consider self-hosted 185 alternatives for mission-critical components. Third, the current deployment has been applied within a single Greek venture capital fund; generalizability across investment stages, sectors, and geographies remains to be established through broader empirical evaluation. Fourth, LLM inference is non-deterministic: repeated runs on identical inputs may produce different

190 synthesis outputs, and the variability of the generated sections is currently unquantified.

6 Data and Workflow Availability

All system artifacts are publicly available in a dedicated repository (Alexandrou, 2026): the n8n JSON workflow file and setup documentation. The repository README details the required credentials and node configuration necessary for replication on a self-hosted or cloud
195 n8n instance.

7 Conclusions and Future Work

Multi-agent orchestration can automate corporate due diligence at a level of reliability previously requiring human analysts. By combining dynamic web retrieval, reverse-engineered endpoint querying, and OCR-based document processing, the pipeline connects unstructured
200 market signals to official financial compliance data in a single automated pass. Structural fallback mechanisms prevent the system from generating unverified financial figures, replacing potential hallucinations with explicit data gaps.

Three directions extend this work. First, integrating the pipeline with the fund’s internal portfolio database would let each report draw on proprietary deal-level data, historical
205 performance metrics, and analyst notes specific to each startup, making the output contextual rather than generic. Second, adding international registries—such as UK Companies House (UK Companies House, 2025) and equivalent European authorities—would extend financial extraction coverage to companies incorporated outside Greece. Third, replacing the third-party PDF extraction service with a self-hosted open-source alternative would reduce operational
210 costs and remove the external dependency without degrading extraction quality.

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